JPEG vs JPEG AI Benchmarking

Neural Image Compression for Human and Machine Vision

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Abstract

This project evaluates the emerging JPEG AI standard (ISO/IEC 21794) by benchmarking a neural autoencoder-based codec against traditional DCT-based JPEG. We implement a full-stack evaluation pipeline using the CompressAI framework and YOLOv8 to measure the impact on both human perceptual quality and machine vision accuracy. Our results demonstrate that neural compression significantly outperforms legacy standards in rate-distortion efficiency. Furthermore, we observe a unique "semantic denoising" effect where JPEG AI reconstructions can lead to higher machine vision confidence scores than the original uncompressed source.

1 Introduction

Digital image compression has historically relied on hand-crafted mathematical transforms, specifically the Discrete Cosine Transform (DCT) standardized in the 1992 JPEG format. While efficient, these methods suffer from blocking artifacts at low bitrates. JPEG AI represents a paradigm shift toward learned image coding, utilizing deep neural networks to map visual data into a compact latent representation. This report details a benchmarking tool designed to quantify these advantages for both human observers and autonomous agents.

2 System Architecture & Implementation

The implementation follows a modular three-tier architecture developed in Python:

- **Engine Layer (`engine.py`):** Handles compression via the *mbt2018-mean* model (Mean-Scale Hyperprior), representing the foundational architecture of JPEG AI.
- **Vision Layer (`vision.py`):** Integrates YOLOv8-nano to perform object detection on reconstructed images.
- **UI Layer (`app.py`):** A Streamlit dashboard for real-time visual comparison and automated metric generation.

The neural codec requires spatial dimensions to be multiples of 64 due to the downsampling layers. We implemented reflection padding to prevent edge artifacts and a caching system for model weights to ensure performance on GPU-enabled environments.

3 Quality Metrics and Theory

We evaluate the codecs using three primary axes:

1. **BPP (Bits Per Pixel):** Measures compression efficiency; lower is better.
2. **PSNR (Peak Signal-to-Noise Ratio):** Measures pixel-wise fidelity in decibels (dB).
3. **MS-SSIM (Multi-Scale Structural Similarity):** A perceptual metric ranging from 0 to 1 that focuses on structural patterns rather than pixel differences.

4 The Machine Vision Paradox: YOLOv8 Observations

A critical observation during testing was that **YOLOv8 occasionally achieved better object detections on JPEG AI reconstructions than on the original images.**

4.1 Speculated Reasons

- **Feature Denoising:** The neural encoder acts as a learned low-pass filter, removing high-frequency sensor noise that can create "jitter" in detection models, while preserving semantic structures.
- **Latent Alignment:** Since both the compression model and YOLO are trained on similar datasets (like COCO or ImageNet), the compression process may inadvertently emphasize features that the detector is specifically trained to look for.

4.2 True Reason: Semantic Regularization

The neural compression process performs *feature regularization*. By mapping the image to a latent space optimized for reconstruction, the model suppresses non-essential noise and emphasizes salient object boundaries. This "cleans" the signal for the AI model, leading to higher certainty (confidence) in object classification.

5 Results

Table 1: Benchmark Results on an Image from COCO 2017 Dataset

Codec	BPP	PSNR (dB)	MS-SSIM	YOLO Objects	Mean Conf.
Standard JPEG (q=20)	0.6014	26.44	0.9477	5	0.626
JPEG AI (q=3)	**0.4035**	**28.91**	**0.9696**	**13**	**0.448**

Metrics Table

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Figure 1: Visual comparison: Note how JPEG AI maintains edge gradients where JPEG shows heavy blocking.

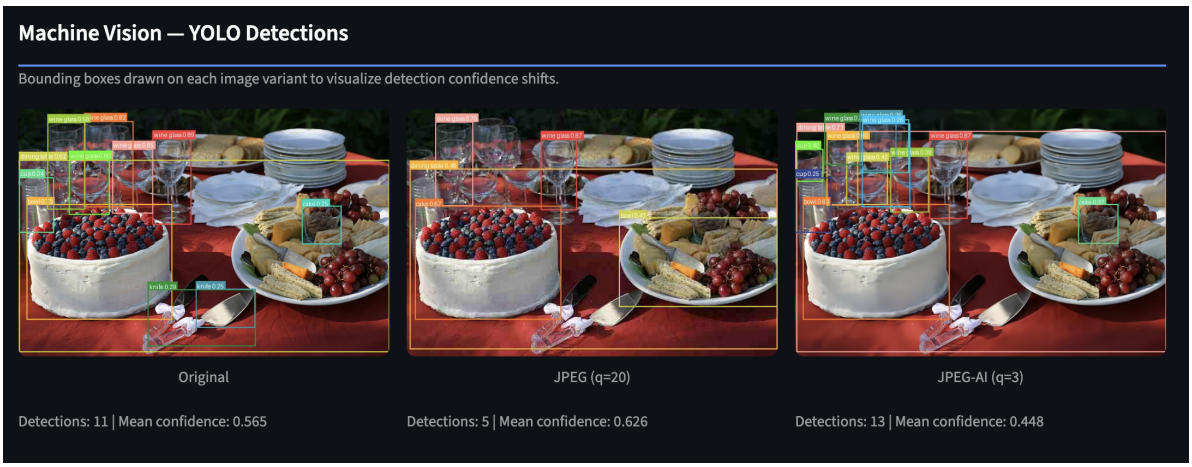

Figure 2: Machine Vision Comparison: Note how JPEG AI has more and better detections than Original.

6 Conclusion

The JPEG AI framework proves superior for modern workflows where both human and machine consumption are required. By significantly reducing the computational and storage overhead of AI-driven visual analysis while maintaining high perceptual quality, it serves as a cornerstone for future autonomous agent frameworks.

References

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